

MOD 309 Course Outline

MOD 309 — Poverty and Household Microsimulation

Course Outline

School	03 Economic Modelling
Level	300 (practitioner / intermediate to advanced)
Hours	40 (indicative band 36–45)
Shape	5 parts · 16 sessions · 9 laboratories · 4 assignments · 1 examination · 1 final project
Delivery	Instructor-led — online, blended or intensive in person
Prerequisites	MOD 303 for the CGE side and MOD 301 for the SAM. Survey statistics to the level of weights, strata and clusters. No prior microsimulation is assumed.
Assessment	Pass 60 %, with 60 % laboratory competency
Version	1.0 · reviewed 2026-08-27 · 12-month cycle

What this course is

A general-equilibrium model tells you what happens to prices, output and factor incomes. It cannot tell you what happens to a household, because it does not contain any. Microsimulation is the bridge: it puts a real distribution of households under the price and income changes a macro model produces, and reports who gains and who loses. This course builds that bridge from both ends — the poverty and incidence measurement on the survey side, and the handover from this school's own CDCGE model — and is exact about which parts of it carry weight and which do not.

Every poverty and inequality figure in this course is computed on the real Utopia National Panel Survey, and every one of them is computed twice: once by the school's own Python and once by daspr, the R port of Araar and Duclos' Distributive Analysis Stata Package, with design-based standard errors. The two agree to nine decimal places on the poverty measures and the Gini, which is what licenses the rest. Thirty-three model interventions are then carried through three thousand households, first as a price pass-through and again with behaviour, and the two are reported side by side. All 51 registered findings are in the appendix.

The rule the course is built on

A rate is a fraction of something. Say what. The same run through the same households moves poverty by nearly a point in either direction depending on the income closure, and a headcount computed per person and per household are different numbers with the same

name. No distributional figure in this course appears without its denominator, its accounting unit and the closure that produced it.

Who it is for

Poverty analysts and social-policy economists who work with household surveys, and modellers who must take a macro result down to households and defend the accounting unit they did it on.

Structure

Part	Title	Sessions	Laboratory	Hours
P1	The survey, its weights and its welfare measure	01–03	01–02	7.1
P2	Poverty: lines, the FGT family, inference, de- composition	04–07	03–04	9.5
P3	Inequality, fiscal incidence and progressivity	08–10	—	4.1
P4	Two findings the data forced	11–12	05–06	6.2
P5	Microsimulation, the handover and reporting Assignment and assessment time	13–16	07–09	11.0
	Total	16	9	40.0

The 16 sessions and 9 laboratories, in teaching order, and the subject matter of each:

P1 The survey, its weights and its welfare measure — Sessions 01–03 · 7.1 h · Laboratory 01–02

- **01 What a Survey Can and Cannot Tell You** — The household survey as an instrument; Sampling: strata, clusters, weights; What a survey observes and what it infers; Synthetic data, and why this course uses it; The standing constraint: no figure here is an estimate; What can still be learned from a file that is not real
- **02 Weights** — Design weights and why they exist; Household weight against person weight; Which one a poverty rate needs; Two fields called population — school gap M-25; Strata and clusters, and what they do to variance; Stating the weight beside every figure

- **03 Welfare Measures** — Consumption against income; Per capita against per adult equivalent; Equivalence scales, and that they are assumptions; Spatial deflation; The ten welfare measures this engine offers; What the choice costs: the same survey, two answers
- *Lab 01 Read the Survey First* — What file is this, and what is its status?
- *Lab 02 Which Weight?* — What does each weight count, and which does a poverty rate need?

P2 Poverty: lines, the FGT family, inference, decomposition — Sessions 04–07 · 9.5 h · Laboratory 03–04

- **04 Poverty Lines** — Absolute, relative and subjective lines; The cost-of-basic-needs approach; International lines and PPP conversion; The eight lines this configuration carries; Relative lines, and why they move with the distribution; Reporting a line rather than assuming it
- **05 The FGT Family** — The general FGT formula; FGT0: how many are poor; FGT1: how far below the line they are; FGT2: how unequal poverty is among the poor; Watts, Sen and Sen-Shorrocks-Thon; Decomposability, and why it matters for policy
- **06 Inference on a Poverty Rate** — Why a survey estimate has sampling variance; Design effects from stratification and clustering; Confidence intervals on FGT measures; Comparing two rates: the overlap fallacy; Small-area estimates, and their limits; How many observations a subgroup needs
- **07 Decomposition** — Population share against poverty contribution; By area, by region; Reading a contribution column; The targeting question a decomposition answers; And the one it does not; Subgroup sample sizes, checked first
- *Lab 03 Poverty, Measured* — How poor is this population, and how sure are you?
- *Lab 04 What the Choices Cost* — How much does the answer depend on decisions nobody sees?

P3 Inequality, fiscal incidence and progressivity — Sessions 08–10 · 4.1 h

- **08 Inequality** — The Lorenz curve and the Gini; Generalised entropy and the Atkinson family; Inequality aversion as an explicit parameter; Percentile ratios and the Palma; Decomposable measures; Which measure to report, and why more than one
- **09 Fiscal Incidence** — Market income to final income; Direct taxes, direct transfers; Indirect taxes and subsidies; In-kind benefits, and what valuing them assumes; The stage sequence as an accounting identity; What incidence analysis assumes about behaviour
- **10 Progressivity** — The concentration coefficient; The Kakwani index for a tax and for a transfer; Reynolds-Smolensky, vertical equity, reranking; Fiscal impoverishment; Net receivers and net payers by quantile; Reading an instrument table

P4 Two findings the data forced — Sessions 11–12 · 6.2 h · Laboratory 05–06

- **11 A Rate That Is Not a Rate** — Every average rate shares one denominator; What that denominator is; Annual household against monthly per-adult-equivalent; A personal income tax at 166 per cent of income; Which columns survive and which do not; The three-part screen for any rate
- **12 A Stage With No Data** — Consumable income, as the framework defines it; The subsidy column, screened; The electricity subsidy, and where it is booked; Two more instruments at exactly zero; What the consumable-income stage actually measures; Reporting a stage under its true name
- *Lab 05 A Rate That Is Not a Rate* — Can this fiscal system take 480 per cent of income?
- *Lab 06 The Stage With No Data* — What does the consumable-income stage actually measure?

P5 Microsimulation, the handover and reporting — Sessions 13–16 · 11.0 h · Laboratory 07–09

- **13 Price-Shock Microsimulation** — Budget shares and the first-order approximation; Own

production, and why it changes the sign; Cost-of-living change per household; Poverty before and after; The distribution of the loss, not just the mean; What a first-order estimate omits

- **14 The CGE Handover** — What a CGE model hands over: prices and factor incomes; The sector-to-commodity-group concordance; The factor-income channel; School gap M-31: the resolution decides what solves; School gap M-09 and M-08: a refusal and a failure are different things; What crosses the boundary, and what it supports once the income closure is stated
- **15 Multidimensional Poverty** — Deprivation indicators and cutoffs; The dual cutoff: deprivation and poverty; Incidence, intensity and the adjusted headcount; Censored headcounts and contributions; Where the two poverty measures disagree; What the choice of indicators decides
- **16 Reporting Poverty Work** — The survey, its vintage and its status; The weight, the welfare measure and the line; Every rate screened before it is quoted; Inference reported, not omitted; How poverty numbers are misused; The poverty note template
- *Lab 07 Who Bears a Price Shock* — A ten per cent food price rise — who pays for it?
- *Lab 08 The Handover* — What actually crossed from the CGE model?
- *Lab 09 The Poverty Note* — What can honestly be said from this survey?

What a graduate can do

1. Explain what a household survey is and what its weights do.
2. Distinguish a household weight from a person weight and say when each is correct.
3. Compute the FGT family and say what each member measures.
4. Attach a standard error to a poverty rate and say why it is necessary.
5. Choose a welfare measure and quantify what the choice costs.
6. Choose a poverty line and say what it is and is not.
7. Compute and read the standard inequality measures.
8. Construct a fiscal incidence analysis through the CEQ income stages.
9. Read progressivity, concentration and Kakwani indices.
10. Screen a rate for unit, period and unit-of-analysis consistency.
11. Run a price-shock microsimulation and report who bears the cost.
12. Describe the CGE-to-microsimulation handover and where it stops.
13. Compute a multidimensional poverty index and say what it adds.
14. Write a poverty note that a reader could act on or refuse.

The formal learning outcomes, and the map from each to the assessment that tests it, are in 12_LMS/learning_outcomes.md.

Assessment

Component	Weight	What it is
Knowledge quizzes	—	A bank of 128 items, used for formative practice and drawn on by the theory paper, which carries the marks

Component	Weight	What it is
Laboratory exercises	35.01 %	9 laboratories, equally weighted. A laboratory is not complete until its validation checks have been read and its named deliverable produced: an irreproducible result is no result
Applied assignments	20 %	The survey and its weights; Poverty, with everything it depends on; Screen before you quote; The poverty note
Theory examination	20 %	Closed-book, drawn from the quiz bank and the review questions
Final project	25 %	A complete poverty and incidence analysis — this material, or a survey the learner is responsible for — delivered as a nine-section poverty note plus the screening scripts that produced it

The per-laboratory weights are an equal split of the laboratory component rounded to two decimals, so the column sums to 100.01 rather than 100. The component weights are the binding ones.

Pass 60 % overall and 60 % laboratory competency. Merit 70 %. Distinction 80 %. One resubmission of a single failed component within eight weeks. A laboratory is not complete until every validation check has been read and the provenance of every figure recorded: an irreproducible result is no result.

Data and access

The survey is real and it is restricted. No laboratory opens a microdata file: what ships is the derived household frame with identifiers, cluster coordinates and anything else disclosive removed, and every estimate carries its unweighted denominator beside its weighted value. Estimates are labelled as estimates and specified quantities as specified; presenting the second as the first is an automatic fail. The course also declines a repair it could make in one line, and says which and why.

Materials and set-up

A 135-page Technical Manual, 17 lecture decks of 207 slides, 9 laboratory decks of 116 slides, a Laboratory Manual, Instructor Guide and Learner Workbook, a 128-item quiz bank, 5 assessment files, 42 datasets, 4 code exercises, 3 case studies and the LMS package. The README lists every file, and COURSE_BUILD_REPORT.md records how each was produced.

The laboratories need Python with pandas, and R with daspr for the distributive analysis. R on this host is installed at C:/Program Files/R/R-4.6.1 but is not on PATH — call Rscript by full path,

and repeat a failed `library()` once, because the Application-Control policy blocks a freshly-touched package DLL on first load only. The behavioural microsimulation needs `TDBmicrosim.jl` and Julia with `-pkgimages=no`.

The 51 findings registered against this school, and the bearing each has on this course, are carried in the Technical Manual's gap appendix and listed in `COURSE_VERSION.json`.

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